

Technology Stack (Architecture & Stack)

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	4 Marks

Technical Architecture

PocketSmart AI follows a layered web architecture. Browser forms are handled by FastAPI routers, validated with Pydantic, processed through recommendation services, sent to Google Gemini when AI generation is required, validated by a deterministic Python budget engine, optionally stored in SQLite through SQLAlchemy, and rendered with Jinja2 templates.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive web pages, planner forms and result cards	HTML5, CSS3, JavaScript, Jinja2
2	Web Backend	HTTP routing, dependency injection and request handling	Python, FastAPI, Uvicorn
3	AI Recommendation Logic	Structured text and multimodal recommendation generation	Google GenAI SDK, Gemini 2.5 Flash Lite
4	Validation	Typed request/response validation	Pydantic
5	Budget Engine	Deterministic subtotals, totals, savings and budget checks	Python
6	Database	Users and recommendation history	SQLite 3, SQLAlchemy ORM
7	Authentication	Password hashing and signed access tokens	bcrypt, PyJWT, HTTP-only cookies
8	Image Processing	Verify and prepare outfit images for multimodal analysis	Pillow
9	Configuration	Environment-based API key and application settings	python-dotenv / Pydantic settings
10	External Search Integration	Creates targeted marketplace/service search URLs	Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, CaratLane search links
11	Testing	Automated unit/integration/end-to-	Pytest

		end testing	
12	Infrastructure	Local ASGI server for current verified build	Uvicorn / local system

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Modular Python web application and ORM	FastAPI, SQLAlchemy, Jinja2, Pydantic, Pillow
2	Security Implementations	Hashed passwords, signed JWT, HTTP-only cookie and environment-held secrets	bcrypt, PyJWT, .env
3	Scalable Architecture	Routers and services are separated so modules can be extended independently	Layered modular FastAPI architecture
4	Availability	Clear failure handling prevents fake recommendations when AI is unavailable	FastAPI exception handling and controlled Gemini failures
5	Performance	Lightweight server-rendered UI; asynchronous web framework; structured AI calls	FastAPI/Uvicorn, Gemini 2.5 Flash Lite